

Lecture 23 – Parametric Polymorphism

COSE212: Programming Languages

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- An **algebraic data type** is a **recursive sum type of product types**.
- ATFAE – TRFAE with **ADTs** and **pattern matching**.
 - **Interpreter** and **Natural Semantics**
 - **Type Checker** and **Typing Rules**
- In this lecture, we will learn **parametric polymorphism**.
- PTFAE – TFAE with **parametric polymorphism**.
 - **Interpreter** and **Natural Semantics**
 - **Type Checker** and **Typing Rules**

1. Parametric Polymorphism
2. PTFAE – TFAE with Parametric Polymorphism
 - Concrete Syntax
 - Abstract Syntax
3. Interpreter and Natural Semantics for PTFAE
4. Type Checker and Typing Rules
 - Type Environment for Type Variables
 - Well-Formedness of Types
 - Function Definition
 - Type Abstraction
 - Type Application
 - Function Application
5. Type Soundness of PTFAE
 - Type Abstraction - Revised

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In the following Scala program, `f` is an identity function and we want to pass ① `1`, ② `true`, and ③ `(y: Int) => y` to `f`, respectively.

```
def f(x: ???): ??? = x;  
f(1); f(true); f((y: Int) => y + 1)
```

Unfortunately, we cannot assign any type to `x` because the type of `x` should be ① `Int`, ② `Boolean`, and ③ `Int => Int`, simultaneously.

How can we resolve this problem? **Polymorphism!**

Polymorphism is to use a single entity as **multiple types**.

There are various kinds of polymorphism:

- **Parametric polymorphism**
- **Subtype polymorphism**
- **Ad-hoc polymorphism** (overloading (Java), type classes (Haskell))
- ...

Among them, let's learn **parametric polymorphism** in this lecture.

Definition (Parametric Polymorphism)

Parametric polymorphism is a form of polymorphism by introducing **type variables** and instantiating them with **type arguments**.

```
def f[T](x: T): T = x;  
f[Int](1); f[Boolean](true); f[Int => Int]((y: Int) => y)
```

The type **T** is a **type variable** (or **type parameter**), and it can be **instantiated** to any types (e.g., `Int`, `Boolean`, and `Int => Int`) by passing them as **type arguments**.

In general, parametric polymorphism is applied to **functions** and **data types**, and they are sometimes called **generic functions** and **generic data types**, respectively.

Many modern typed languages support **parametric polymorphism**:

- Scala

```
def f[T](x: T): T = x
```

- C++

```
template <typename T> T f(T x) { return x; }
```

- Rust

```
fn f<T>(x: T) -> T { return x; }
```

- Haskell

```
f :: a -> a  
f x = x
```

- ...

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Let's extend TFAE into PTFAE to support **parametric polymorphism** for any expressions rather than only functions or data types.

```
/* PTFAE */  
val f = forall[T] { (x: T) => x }           // [T] (T => T)  
val x = f[Number](42)                     // Number  
val y = f[Number => Number](f[Number])    // Number => Number  
val z = f[[T] (T => T)](f)                 // [T] (T => T)  
...
```

`forall[t]` e parameterizes an expression e with a type variable t, and `e[t]` instantiates the type variable with a type t of an expression e.

For PTFAE, we need to extend **expressions** of TFAE with

- 1 **type abstraction** (`forall`)
- 2 **type application**
- 3 **type variable**
- 4 **polymorphic type**

For PTFAE, we need to extend **expressions** of TFAE with

- 1 **type abstraction** (forall)
- 2 **type application**
- 3 **type variable**
- 4 **polymorphic type**

We can extend the **concrete syntax** of TFAE as follows:

```
// expressions
<expr> ::= ...
        | "forall" "[" <id> "]" <expr>
        | <expr> "[" <type> "]"

// types
<type> ::= ...
        | <id>
        | "[" <id> "]" <type>
```

Expressions	$\mathbb{E} \ni e ::= \dots$
	$\quad \mid \forall \alpha. e \quad (\text{TypeAbs})$
	$\quad \mid e[\tau] \quad (\text{TypeApp})$
Types	$\mathbb{T} \ni \tau ::= \dots$
	$\quad \mid \alpha \quad (\text{VarT})$
	$\quad \mid \forall \alpha. \tau \quad (\text{PolyT})$
Type Variables	$\alpha \in \mathbb{X}_\alpha \quad (\text{String})$

```
enum Expr:
  ...
  case TypeAbs(name: String, body: Expr)
  case TypeApp(expr: Expr, ty: Type)

enum Type:
  ...
  case VarT(name: String)
  case PolyT(name: String, ty: Type)
```

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For PTFAE, we need to 1) implement the **interpreter** with environments:

```
def interp(expr: Expr, env: Env): Value = ???
```

and 2) define the **natural semantics** with environments:

$$\sigma \vdash e \Rightarrow v$$

with a new kind of values called **type abstraction values** to **delay** the evaluation of type abstractions until they are applied to type arguments.

Values $\mathbb{V} \ni v ::= n$ (NumV)
 | $\langle \lambda x.e, \sigma \rangle$ (CloV)
 | $\langle \forall \alpha.e, \sigma \rangle$ (TypeAbsV)

```
enum Value:  
  case NumV(number: BigInt)  
  case CloV(param: String, body: Expr, env: Env)  
  case TypeAbsV(name: String, body: Expr, env: Env)
```

```
def interp(expr: Expr, env: Env): Value = expr match
  ...

  case TypeAbs(name, body) => TypeAbsV(name, body, env)

  case TypeApp(expr, ty) => interp(expr, env) match
    case TypeAbsV(name, body, fenv) => interp(body, fenv)
    case v => error(s"not a type abstraction: ${v.str}")
```

$$\boxed{\sigma \vdash e \Rightarrow v}$$

$$\text{TypeAbs} \frac{}{\sigma \vdash \forall \alpha. e \Rightarrow \langle \forall \alpha. e, \sigma \rangle}$$

$$\text{TypeApp} \frac{\sigma \vdash e \Rightarrow \langle \forall \alpha. e', \sigma' \rangle \quad \sigma' \vdash e' \Rightarrow v}{\sigma \vdash e[\tau] \Rightarrow v}$$

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Let's ❶ design **typing rules** of PTFAE to define when an expression is well-typed in the form of:

$$\Gamma \vdash e : \tau$$

and ❷ implement a **type checker** in Scala according to typing rules:

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = ???
```

The type checker returns the **type** of e if it is well-typed, or rejects it and throws a **type error** otherwise.

Similar to TFAE, we will keep track of the **variable types** using a **type environment** Γ as a mapping from variable names to their types.

$$\text{Type Environments} \quad \Gamma \in \mathbb{X} \xrightarrow{\text{fin}} \mathbb{T} \quad (\text{TypeEnv})$$

```
type TypeEnv = Map[String, Type]
```


However, we need additional information in type environments to keep track of which **type variables** are defined by **type abstractions**.

Type Environments $\Gamma \in (\mathbb{X} \xrightarrow{\text{fin}} \mathbb{T}) \times \mathcal{P}(\mathbb{X}_\alpha)$ (TypeEnv)

We write $\alpha \in \text{dom}(\Gamma)$ when the type variable α is defined in Γ , and $\Gamma[\alpha]$ for an extension of Γ with α defined.

```
case class TypeEnv(  
  vars: Map[String, Type] = Map(),  
  tys: Set[String] = Set(),  
) {  
  def addVar(pair: (String, Type)): TypeEnv =  
    TypeEnv(vars + pair, tys)  
  def addVars(pairs: Iterable[(String, Type)]): TypeEnv =  
    TypeEnv(vars ++ pairs, tys)  
  def addType(name: String): TypeEnv = TypeEnv(vars, tys + name)  
}
```

Similar to ATFAE, we need to check the **well-formedness** of types with **type environment** to prevent the use of not-defined type variables.

$$\boxed{\Gamma \vdash \tau}$$

$$\frac{}{\Gamma \vdash \text{num}} \quad \frac{\Gamma \vdash \tau \quad \Gamma \vdash \tau'}{\Gamma \vdash \tau \rightarrow \tau'} \quad \frac{\alpha \in \text{dom}(\Gamma)}{\Gamma \vdash \alpha} \quad \frac{\Gamma[\alpha] \vdash \tau}{\Gamma \vdash \forall \alpha. \tau}$$

```
def mustValid(ty: Type, tenv: TypeEnv): Type = ty match
  case NumT =>
    NumT
  case ArrowT(pty, rty) =>
    ArrowT(mustValid(pty, tenv), mustValid(rty, tenv))
  case VarT(name) =>
    if (!tenv.tys.contains(name)) error(s"unknown type: $name")
    VarT(name)
  case PolyT(name, ty) =>
    PolyT(name, mustValid(ty, tenv.addType(name)))
```

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case Fun(param, paramTy, body) =>
  mustValid(paramTy, tenv)
  ArrowT(paramTy, typeCheck(body, tenv.addVar(param -> paramTy)))
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-Fun} \frac{\Gamma \vdash \tau \quad \Gamma[x : \tau] \vdash e : \tau'}{\Gamma \vdash \lambda x : \tau. e : \tau \rightarrow \tau'}$$

Similar to ATFAE, we check the **well-formedness** of parameter types.

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeAbs(name, body) =>
  ???
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeAbs} \frac{\text{???}}{\Gamma \vdash \forall \alpha. e : \text{???}}$$

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeAbs(name, body) =>
  typeCheck(body, ???)
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeAbs} \frac{??? \vdash e : ???}{\Gamma \vdash \forall \alpha. e : ???}$$

First, we need to check the **body** of a type abstraction.

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeAbs(name, body) =>
  typeCheck(body, tenv.addType(name))
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeAbs} \frac{\Gamma[\alpha] \vdash e : \tau}{\Gamma \vdash \forall \alpha. e : ???}$$

We need to extend the **type environment** with the type variable α .

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeAbs(name, body) =>
  PolyT(name, typeCheck(body, tenv.addType(name)))
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeAbs} \frac{\Gamma[\alpha] \vdash e : \tau}{\Gamma \vdash \forall \alpha. e : \forall \alpha. \tau}$$

The type of a type abstraction is a **polymorphic type**.

It is indeed **type unsound**, and we will fix it later in this lecture.

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeApp(expr, ty) => ???
```

$$\boxed{\Gamma \vdash e : \tau}$$
$$\tau\text{-TypeApp} \frac{\text{???}}{\Gamma \vdash e[\tau] : \text{???}}$$


```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeApp(expr, ty) => typeCheck(expr, tenv); ???
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeApp} \frac{\Gamma \vdash e : ???}{\Gamma \vdash e[\tau] : ???}$$

First, we need to check the type of e with the given type environment Γ .

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeApp(expr, ty) => typeCheck(expr, tenv) match
  case PolyT(name, bodyTy) => ???
  case t => error(s"not a polymorphic type: ${t.str}")
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeApp} \frac{\Gamma \vdash e : \forall\alpha.\tau' \quad ???}{\Gamma \vdash e[\tau] : ???}$$

But, we need to allow type application only if the type of e is a **polymorphic type**.

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeApp(expr, ty) => typeCheck(expr, tenv) match
  case PolyT(name, bodyTy) => mustValid(ty, tenv); ???
  case t => error(s"not a polymorphic type: ${t.str}")
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeApp} \frac{\Gamma \vdash e : \forall\alpha.\tau' \quad \Gamma \vdash \tau}{\Gamma \vdash e[\tau] : ???}$$

We also need to check the **well-formedness** of type argument τ .

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeApp(expr, ty) => typeCheck(expr, tenv) match
  case PolyT(name, bodyTy) => subst(bodyTy, name, mustValid(ty, tenv))
  case t => error(s"not a polymorphic type: ${t.str}")
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeApp} \frac{\Gamma \vdash e : \forall \alpha. \tau' \quad \Gamma \vdash \tau}{\Gamma \vdash e[\tau] : \tau'[\alpha \leftarrow \tau]}$$

Finally, we need to **substitute** the type variable α with the type argument τ in the body type τ' .

$\tau'[\alpha \leftarrow \tau]$ means replacing all occurrences of **free type variable** α in τ' with τ . For example,

$$(\alpha \rightarrow \beta \rightarrow (\forall \alpha. \alpha) \rightarrow \alpha)[\alpha \leftarrow \text{num}] = \text{num} \rightarrow \beta \rightarrow (\forall \alpha. \alpha) \rightarrow \text{num}$$

We can implement the substitution as follows:

```
def subst(bodyTy: Type, name: String, ty: Type): Type = bodyTy match
  case NumT => NumT
  case ArrowT(pty, rty) =>
    ArrowT(subst(pty, name, ty), subst(rty, name, ty))
  case VarT(x) =>
    if (name == x) ty
    else VarT(x)
  case PolyT(x, bodyTy) =>
    if (name == x) PolyT(x, bodyTy)
    else PolyT(x, subst(bodyTy, name, ty))
```

A real implementation needs **capture-avoiding** substitution: x must be renamed to a fresh type variable when it is free in ty .
Now, we can instantiate type variables with given types in specific types:

```
val ty = Type("T => U => ([T](T)) => T")
subst(ty, "T", NumT).str           // "Number => U => ([T](T)) => Number"
```

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case App(fun, arg) => typeCheck(fun, tenv) match
  case ArrowT(paramTy, retTy) =>
    mustSame(typeCheck(arg, tenv), paramTy)
    retTy
  case t => error(s"not a function type: ${t.str}")
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-App} \frac{\Gamma \vdash e_0 : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash e_1 : \tau_1}{\Gamma \vdash e_0(e_1) : \tau_2}$$

While we can use the same rule in TFAE, but we can improve it.

```
/* PTFAE */
val f = ((x: [T](T => T)) => x[Number](7)) // ([T](T => T)) => Number
val x = forall[U] (x: U) => x // [U] (U => U)
f(x) // type checking failed: [T](T => T) != [U](U => U)
```

Let's define the equivalence ($\equiv$) of types as follows:

```
def isSame(lty: Type, rty: Type): Boolean = (lty, rty) match
  case (NumT, NumT) => true
  case (ArrowT(lpty, lrty), ArrowT(rpty, rrtty)) =>
    isSame(lpty, rpty) && isSame(lrty, rrtty)
  case (VarT(lname), VarT(rname)) => lname == rname
  case (PolyT(lname, lty), PolyT(rname, rty)) =>
    isSame(lty, subst(rty, rname, VarT(lname)))
  case _ => false

def mustSame(l: Type, r: Type): Unit =
  if (!isSame(l, r)) error(s"type mismatch: ${l.str} != ${r.str}")
```

$$\boxed{\tau \equiv \tau}$$

$$\frac{}{\text{num} \equiv \text{num}} \quad \frac{\tau_1 \equiv \tau'_1 \quad \tau_2 \equiv \tau'_2}{(\tau_1 \rightarrow \tau_2) \equiv (\tau'_1 \rightarrow \tau'_2)} \quad \frac{}{\alpha \equiv \alpha} \quad \frac{\tau \equiv \tau'[\alpha' \leftarrow \alpha]}{\forall \alpha. \tau \equiv \forall \alpha'. \tau'}$$

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case App(fun, arg) => typeCheck(fun, tenv) match
  case ArrowT(paramTy, retTy) =>
    mustSame(typeCheck(arg, tenv), paramTy)
    retTy
  case t => error(s"not a function type: ${t.str}")
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-App} \frac{\Gamma \vdash e_0 : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash e_1 : \tau_3 \quad \tau_1 \equiv \tau_3}{\Gamma \vdash e_0(e_1) : \tau_2}$$

While we can use the same rule in TFAE, but we can improve it.

```
/* PTFAE */
val f = ((x: [T](T => T)) => x[Number](7)) // ([T](T => T)) => Number
val x = forall[U] (x: U) => x // [U] (U => U)
f(x) // well-typed: [T](T => T) == [U](U => U)
```


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Definition (Type Soundness)

A **type system** is **sound** if it guarantees that a **well-typed** program will **never** cause a **type error** at run-time.

```
/* PTFAE */  
val f = forall[U] (x: U) => {  
    val y = forall[U] x           // [U] (U)  
    y[Number => Number]           // Number => Number  
}                                  // [U] (U => (Number => Number))  
val g = f[Number](1)             // Number => Number  
g(2)                             // Number
```

It throws a **type error** when evaluating `1(2)` at run-time while this expression is **well-typed** (i.e., **unsound type system**).

We can resolve this problem by **forbidding** the redefinition of **same type variable** in the scope of **type abstractions**!

```
def typeCheck(expr: Expr, tenv: TypeEnv): Type = expr match
...
case TypeAbs(name, body) =>
  if (tenv.tys.contains(name)) error(s"already defined type: $name")
  PolyT(name, typeCheck(body, tenv.addType(name)))
```

$$\boxed{\Gamma \vdash e : \tau}$$

$$\tau\text{-TypeAbs} \frac{\alpha \notin \text{dom}(\Gamma) \quad \Gamma[\alpha] \vdash e : \tau}{\Gamma \vdash \forall \alpha. e : \forall \alpha. \tau}$$

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<https://github.com/ku-plrg-classroom/docs/tree/main/cose212/ptfae>

- Please see above document on GitHub:
 - Implement `typeCheck` function.
 - Implement `interp` function.
- It is just an exercise, and you **don't need to submit** anything.
- However, some exam questions might be related to this exercise, so we **strongly recommend** solving it by yourself.

The MiniScala language is a toy language inspired by Scala.¹

<https://github.com/ku-plrg-classroom/docs/tree/main/cose212/mini-scala>

- Please see above document on GitHub:
 - Implement `typeCheck` function.
 - Implement `interp` functions.
- It is just a project, and you **don't need to submit** anything.
- However, some exam questions might be related to this project, so we **strongly recommend** solving it by yourself.
- The given test cases are **not sufficient**. Please write your own test cases to test your implementation.

¹<https://www.scala-lang.org/>

- Subtype Polymorphism

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